

Problem Statement

Hotel Room Booking Management System

Background

A hotel wants to digitize its room booking process.

The management needs a backend application to store and manage customer booking information.

The application should be developed using:

- FastAPI
- PostgreSQL

And Demonstrate

- CRUD APIs
- PostgreSQL Integration
- SQLAlchemy ORM
- Pydantic Models
- Response Models
- Status Codes
- Exception Handling
- Dependency Injection
- Middleware

Database Details

Database Name

hotel_db

Table Name

HotelBooking

Columns

Column Name	Data Type
booking_id	Integer
customer_name	String
room_type	String
number_of_days	Integer

Tasks to Perform

Task 1

Create PostgreSQL database.

Task 2

Create:

- database.py
- models.py
- schemas.py
- main.py

Task 3

Create SQLAlchemy model.

Task 4

Create Pydantic model for:

Create Booking

Task 5

Create Response Model.

Task 6

Create POST API

Requirements:

- Add booking details
- Return response model
- Return appropriate status code

Task 7

Create GET API

Requirements:

- Fetch all bookings
- Use response model
- Return status code

Task 8

Create PUT API

Requirements:

- Update room type
- Update number of days
- Use path parameter

Task 9

Create DELETE API

Requirements:

- Delete booking using booking_id

Task 10

Implement Exception Handling

Cases:

Booking not found

Duplicate Booking ID

Use:

- HTTPException
- Appropriate status codes

Task 11

Implement Dependency Injection

Use:

get_db()

Depends()

instead of global session.

Task 12

Implement Middleware

Log:

Request Method

Request URL

Processing Time